

Hikari Murayama

Academic appointments

University of Washington, Seattle, WA September 2026 – Present
Postdoctoral Fellow, Department of Atmospheric and Climate Science

Education

University of California, Berkeley, Berkeley, CA May 2021 – May 2026
Ph.D. in Energy and Resources
Visiting student, Stanford Doerr School of Sustainability (July 2024 – July 2026)
Dissertation: *Enhancing our knowledge of the carbon budget: past and present*

University of California, Berkeley, Berkeley, CA August 2019 – May 2021
M.S. in Energy and Resources

Wellesley College, Wellesley, MA August 2012 – May 2016
Bachelor of Arts in Chemical Physics
Cum Laude (GPA: 3.71/4.00), Departmental Honors, Sigma Xi
Honors Thesis: *DNA G-Quadruplex Formation in the Bdellovibrio bacteriovorus Genome: An in vitro Study Exploring Temperature, Time, and Crowding*

Peer Reviewed Publications

Murayama, H., Noda, E., Chong, T., Druckenmiller, H., Ferguson, J., Greenhill, S., Hsiang, S., Ilin, C., Kee, T., Madestam, A., Nordfors, N., Tompsett, A., Wang, S.. *Developing high resolution, historical land cover maps in Africa using deep learning and an aerial photography archive*. 2026 December. *In Preparation*

Murayama, H., Wang, S., Sherman, L., Cohen, R., Hsiang, S. *Parameterizing deep learning for carbon dioxide emission quantification for power plants*. 2026 October. *In Preparation*

Proctor, J., Carleton, T., Chong, T., Fransen, T., Greenhill, S., Katz, J., **Murayama, H.**, Sherman, L., Tseng, J., Druckenmiller, H., Hsiang, S.. *A global evaluation of machine learning with satellite imagery for measuring ground conditions?* 2026 May. *In Review*

Baldassare, B., Lute, A., **Murayama, H.**, Kingdon, C., Schwalm, C.. *Even Small Companies Can Save Lives by Reducing Emissions*. 2026 May. *In Review*

Brumberg, H., Dee, L., **Murayama, H.**, Barrientos, J.J.A, Bessesen, B., Bouffard, M.G., Burgess, M.G., Cortés, J., Furey, S., Hernández, N., Luger, A.M., Madden, M., Pauline, E., Schmitt, R.J.P., Siegel, K.J., Vargas-Araya, L., Whitworth, A., Newton, P.. *Riparian vegetation reduces coastal turbidity*. 2026 February. In: Communications Sustainability

Brumberg, H., Furey, S., Bouffard, M.G., Mata Quirós, M.J., **Murayama, H.**, Neyestani, S., Pauline, E., Whitworth, A., Madden, M. *Increasing forest cover and connectivity both inside and outside of Protected Areas in southwestern Costa Rica*. 2024 March. In: Remote Sensing.

Decter-Frain, A., Sachdeva, P., Collingwood, L., Burke, J., **Murayama, H.**, Barreto, M., ... Zingher, J. *Comparing Methods for Estimating Demographics in Racially Polarized Voting Analyses*. 2023 August. In: Sociological Methods & Research

Tsao, L.H., Shepardson-Fungairiño, S., **Murayama, H.**, Cecere, A., Wren, E., Núñez, M.. *Assessing the Potential for DNA Quadruplex Formation in the Predatory Bacterium Bdellovibrio bacteriovorus*. 2022 September. In: Biochemistry.

Figuerola, C., **Murayama, H.**, Amorim, P.C., White, A., Quiteri, A., Luo, T., Aguilera, A., Smith, A.D.R., Lyles, C.R., Robinson, V., Von Vacano, C. *Applying the Digital Health Social Justice Guide*. 2022 March. In: Frontiers in Digital Health.

Fellowships and Grants

- 2026 - Gordon and Betty Moore Foundation – Postdoctoral fellowship
- 2026 - UC Berkeley Graduate Division – Conference Travel Grants
- 2025 - Quad Fellowship by Institute of International Education – Senior Fellow Ambassador
- 2025 - University of California Dissertation-Year Fellowship – Fellow
- 2024 - UC Berkeley Energy and Resources Group – Mini grant
- 2024 - Quad Fellowship by Institute of International Education – Fellow

Last updated: September 2026

- 2024 - UC Berkeley Energy Institute at Haas and the Opportunity Lab: Energy and Environmental Economics Mentoring Program – Mentor
- 2022 - Google Cloud Education Programs – Google Cloud Research Credits
- 2021 - UC Berkeley D-Lab – Senior Data Science Fellow
- 2021 - UC Berkeley Global Policy Lab – Doctoral Fellow
- 2020 - UC Berkeley D-Lab – PIT-UN Fellow and Data Science Fellow
- 2020 - University of Washington eScience Institute – Data Science for Social Good Fellowship
- 2019 - UC Berkeley Graduate Division – Conference Travel Grants

Awards

American Geophysical Union

- 2024 - Outstanding Student Presentation Award in Atmospheric Chemistry at AGU Fall Meeting 2024

Energy and Resources Group at University of California, Berkeley

- 2021 - Kay Burns Advising and Service Award (*awarded to group: Master's Seminar Series Task Force*)

American Society for Photogrammetry and Remote Sensing

- 2019 - 3rd Place in Student Oral Presentations at Pecora Symposium 2019

Wellesley College

- 2016 - Phyllis Fleming Prize for Excellence in Physics
- 2013 - First-Year Chemistry Award

Research Experience

Global Policy Lab at University of California, Berkeley, Berkeley, CA Principle Investigator: Solomon Hsiang, PhD Emissions detection using remote sensing and deep learning Using historical aerial photography for land cover maps	January 2021 – Present
Lawrence Berkeley National Laboratory, Berkeley, CA Principle Investigator: Daniel Feldman, PhD Air pollution pattern changes with COVID-19 Shelter in Place orders Remote sensing for plastic debris quantification	March 2020 – May 2022
D-Lab at University of California, Berkeley, Berkeley, CA Principle Investigator: Claudia von Vacano, PhD Digital health social justice	January 2020 – May 2021
eScience Institute at the University of Washington, Seattle, WA Principle Investigator: Matt Barreto, PhD and Loren Collingwood, PhD Voting rights dilution	June 2020 – August 2020
Wellesley College Chemistry Department, Wellesley, MA Principle Investigator: Megan E. Nuñez, PhD. DNA G-Quadruplex formation using Nuclear Magnetic Resonance (NMR) and Circular Dichromatism	September 2014 – May 2016
University of California, Berkeley Chemical Engineering Department, Berkeley, CA Principle Investigator: Jeffrey Reimer, PhD Nitrogen vacancy centers in diamonds using NMR and Electron Paramagnetic Resonance	June – August 2014
Wellesley College Physics Department, Wellesley, MA Principle Investigator: James Battat, PhD Cosmic ray muon detection	September 2013 – May 2014
California Department of Toxic Substances Control, Berkeley, CA Principle Investigator: Sabrina Crispo Smith, PhD Organophosphate flame-retardants in dust and dirt samples	June – August 2013

Academic Service

Energy and Resources Group at University of California, Berkeley

Hiring committee for Energy and Resources Group Professor

October – December 2022

Last updated: September 2026

Geospatial Innovation Facility (GIF) at University of California, Berkeley

Hiring committee for web developer

June – August 2022

Teaching Experience**University of California, Berkeley, Berkeley, CA**

PP275 – Spatial Data Analysis (TA)

Fall 2022

D-Lab – Coding in R workshop (Instructor)

Spring 2020 – Spring 2022

Coding in Python workshop (Instructor)

Data for Housing workshop (*co-developed*)Digital Health Social Justice workshop (Instructor, *co-developed*)Geospatial Analysis in R workshop (Instructor, *co-developed*)Geospatial Analysis in Python workshop (Instructor, *co-developed*)Geospatial Analysis in QGIS workshop (Instructor, *co-developed*)

ENERES176 – Climate Change Economics (TA)

Fall 2019

Mentorship Experience**Getting into Graduate School (GiGS) at University of California, Berkeley**

August 2025 – May 2026

Mentor

Mentored one undergraduate student in their journey towards applying to graduate school

Data Discovery Program at University of California, Berkeley

August 2025 – May 2026

Mentor

Lead and mentored three undergraduates in research for updating a deep learning pipeline to quantify carbon dioxide emissions from powerplants using a simulated dataset

Environmental and Energy Economics Mentoring Program at University of California, Berkeley August 2024 – May 2025Mentor

Lead and mentored two undergraduates in conducting research in quantifying tropical cyclone impacts in Madagascar

Volunteer Experience**Ready&, Tokyo, Japan**

September 2015 – May 2022

Co-founder, Advisor (2016-2022), Director (2015-2016)

Grassroots, non-profit initiative run by college students to empower Japanese high school girls throughout Japan.

Sponsored by American Center Japan, Harvard Club of Japan, Japan-U.S. Friendship Commission among others.

Girls Who Code, Washington D.C.

August 2017 – May 2018

Facilitator

Weekly meetings for 2 hours teaching middle and high school girls how to code in Scratch, Python, and Java.

Work and Professional Experience**NASA DEVELOP National Program, Athens, GA**

August 2018 – April 2019

Assistant Center Lead, Project Lead**Bates White Economic Consulting, Washington D.C.**

September 2016 – August 2018

Consultant II (2017-2018), Consultant I (2016-2017)**AIG, Boston, MA**

June 2015 – August 2015

Risk Engineer/Loss Prevention Consultant**Conference Presentations**

Murayama, H., Chong, T., Druckenmiller, H., Ferguson, J., Greenhill, S., Hsiang, S., Ilin, C., Kee, T., Madestam, A., Noda, E., Nordfors, N., Tompsett, A., Wang, S.. Using One Million Aerial Photographs to Quantify Historical Land Cover Changes in Africa In: American Geophysical Union Fall Meeting 2025. 2025 December. New Orleans, LA.

Ikarashi, Y., Li, V., **Murayama, H.**, Sawada, A., Tanaka, H.. *Barriers to reaching potentials: an investigation into Japan's workforce and hope for innovation* [Oral Presentation] In: Quad Fellowship Spring Symposium 2025. 2025 May. Remote.

Murayama, H., Chong, T., Druckenmiller, H., Ferguson, J., Greenhill, S., Hsiang, S., Ilin, C., Kee, T., Madestam, A., Noda, E., Nordfors, N., Tompsett, A., Wang, S.. *Using untapped aerial photography archives to quantify historical forest cover changes in Africa* [Oral Presentation] In: Sustainability Data Science Conference. 2024 April. Palo Alto, CA.

Murayama, H., Wang, S., Sherman, L., Cohen, R.C., Hsiang, S.M.. *Toward a generalizable deep learning approach to estimate CO₂ emissions from power plants* [Oral Presentation] In: American Geophysical Union Fall Meeting 2024. 2024 December. Washington, DC.

Murayama, H., Wang, S., Sherman, L., Cohen, R.C., Hsiang, S.M.. *Quantifying CO₂ emissions using deep learning and remote sensing* [Oral Presentation] In: Graduate Climate Conference 2024. 2024 November. Seattle, WA.

Murayama, H., Chong, T., Druckenmiller, H., Ferguson, J., Greenhill, S., Hsiang, S., Ilin, C., Kee, T., Madestam, A., Noda, E., Nordfors, N., Tompsett, A., Wang, S.. *Developing high resolution, historical land cover maps in Africa using deep learning and an aerial photography archive* [Oral Presentation] In: American Geophysical Union Fall Meeting 2023. 2023 December. San Francisco, CA.

Murayama, H., Wang, S., Cohen, R.C., Hsiang, S.M.. *Measuring CO₂ emissions using deep learning and remote sensing* [Poster] In: American Geophysical Union Fall Meeting 2022. 2022 December. Chicago, IL.

Murayama, H., Carleton, T., Chong, T., Fransen, T., Greenhill, S., Katz, J., Proctor, J., Sherman, L., Tseng, J., Druckenmiller, H., Hsiang, S.. *Mapping over 100 Variables using an image of Earth*. [Oral Presentation] In: The Workshop for Environmental Economics and Data Science. 2022 October. Eugene, OR.

Burke, J., Decter-Frain, A., **Murayama, H.**, Sachdeva, P. *eiCompare: Making Every Vote Count*. [Oral Presentation] In: Learning and Doing Data for Good. 2022 September. Seattle, WA.

von Vacano, C., **Murayama, H.**, Olojo, S. *Developing Critical Frameworks for Analyzing STEM Program Structures* [Oral Presentation] In: 2021 National Workshop on Data Science Education. 2021 June. Remote.

Bouffard, M., **Murayama, H.**, Furey, S., Bartlett, B., Palmer, R., Bouffard, M., Ingram, S., and Madden, M. *Determining Habitat Suitability to Establish a Jaguar Corridor between the Talamanca Mountains and the Osa Peninsula in Costa Rica*. [Oral Presentation] In: ASPRS 2021 Annual Conference. 2021 April. Remote.

Barreto, M., Burke, J., Collingwood, L., Decter-Frain, A., **Murayama, H.**, Sachdeva, P. *eiCompare: Comparing BISG to CVAP Estimates in Racially Polarized Voting Analyses*. [Oral Presentation] In: Politics of Race, Immigration, & Ethnicity Consortium. 2020 November. Newark, DE.

Murayama, H., Furey, S., Bartlett, B., Palmer, R., Bouffard, M., Ingram, S., and Madden, M. *Geospatial Modeling of Human-Wildlife Conflict and Habitat Suitability for Jaguar Corridors in Costa Rica*. [Oral Presentation] In: PECORA 21/ISRSE 38 Earth Observation – Continuous Monitoring of Our Changing Planet: From Sensors to Decisions. 2019 October. Baltimore, MD.

Murayama, H., Furey, S., Bartlett, B., Palmer, R., Bouffard, M., Ingram, S., and Madden, M. *Determining Habitat Suitability to Establish a Jaguar Corridor between the Talamanca Mountains and the Osa Peninsula in Costa Rica*. [Oral Presentation] In: NASA DEVELOP Southeast Spring Closeout at NOAA National Center for Environmental Information. 2019 April. Asheville, NC.

Murayama, H., Furey, S., Barney, M., Neyestani, S., Bouffard, M., and Madden, M. *Evaluating Potential Sites for Coral Reef Rehabilitation in the Golfo Dulce, Costa Rica Based on Turbidity and Sea Surface Temperature*. [Oral Presentation] In: NASA DEVELOP Southeast Fall Closeout at University of Georgia. 2018 December. Athens, GA.

Skills

Programming

Python, R, Git/GitHub, STATA, MATLAB

Geospatial Analysis

Google Earth Engine, Python, R, QGIS

Languages

Japanese

Professional Affiliations

American Geophysical Union, Association for Women in Science, American Society for Photogrammetry and Remote Sensing

References

Solomon Hsiang, Ph.D.

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Ronald C. Cohen, Ph.D.

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Data, Systems, and Society*
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